

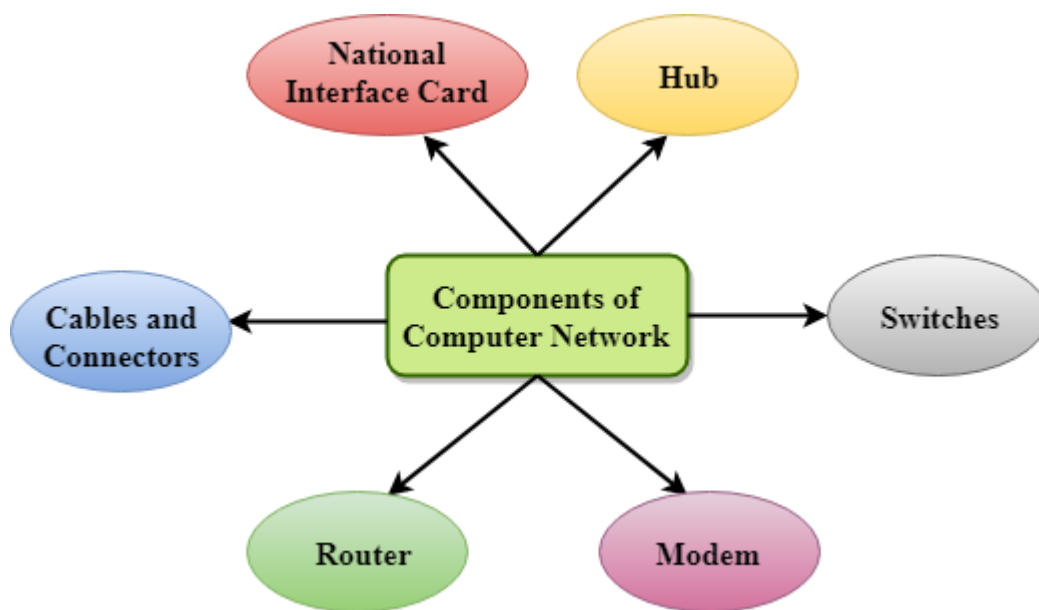
Unit I: Introduction to Computer Networks

Introduction to Networks:

What is a Computer Network?

- Computer Network is a group of computers connected with each other through wires, optical fibres or optical links so that various devices can interact with each other through a network.
- The aim of the computer network is the sharing of resources among various devices.

Components of Computer Network (Hardware)



Major components of a computer network are:

NIC (National interface card)

NIC is a device that helps the computer to communicate with another device. The network interface card contains the hardware addresses, the data-link layer protocol use this address to identify the system on the network so that it transfers the data to the correct destination.

There are two types of NIC: wireless NIC and wired NIC.

- **Wireless NIC:** All the modern laptops use the wireless NIC. In Wireless NIC, a connection is made using the antenna that employs the **radio wave technology**.
- **Wired NIC:** Cables use the **wired NIC** to transfer the data over the medium.

Hub

Hub is a central device that splits the network connection into multiple devices. When computer requests for information from a computer, it sends the request to the Hub. Hub distributes this request to all the interconnected computers.

Switches

Switch is a networking device that groups all the devices over the network to transfer the data to another device. A switch is better than Hub as it does not broadcast the message over the network, i.e., it sends the message to the device for which it belongs to. Therefore, we can say that switch sends the message directly from source to the destination.

Cables and connectors

Cable is a transmission media that transmits the communication signals. **There are three types of cables:**

- **Twisted pair cable:** It is a high-speed cable that transmits the data over **1Gbps** or more.
- **Coaxial cable:** Coaxial cable resembles like a TV installation cable. Coaxial cable is more expensive than twisted pair cable, but it provides the high data transmission speed.
- **Fibre optic cable:** Fibre optic cable is a high-speed cable that transmits the data using light beams. It provides high data transmission speed as compared to other cables. It is more expensive as compared to other cables, so it is installed at the government level.

Router

Router is a device that connects the LAN to the internet. The router is mainly used to connect the distinct networks or connect the internet to multiple computers.

Modem

Modem connects the computer to the internet over the existing telephone line. A modem is not integrated with the computer motherboard. A modem is a separate part on the PC slot found on the motherboard.

Application of Computer Network

There are a variety of fields in computer networks that are used in industries. Some of them are as follows:

1. Internet and World Wide Web

In computer networks, we have a global internet, also known as the World Wide Web, that offers us various features like access to websites, online services and retrieval of information.

With the help of the World Wide Web, we can browse, and we can do search, and access web pages and multimedia content.

2. Communication

With the help of computer networks, communication is also easy because we can do email, instant messaging, voice and video calls and video conferencing, which helps us to communicate with each other effectively. People can use these features in their businesses and organizations to stay connected with each other.

3. File Sharing and Data Transfer

Data transfer and file sharing are made possible by networks that connect different devices. This covers file sharing within a business setting, file sharing between personal devices, and downloading/uploading of content from the internet.

4. Online gaming

Multiplayer online games use computer networks to link players from all over the world, enabling online competitions and real-time gaming experiences.

5. Remote Access and Control

Networks enable users to access and control systems and devices from a distance. This is helpful when accessing home automation systems, managing servers, and providing remote IT support.

6. Social media

With the help of a computer network, we can use social media sites like Facebook, Twitter and Instagram to help people set up their profiles, and we can connect with others and share content on social media.

7. Cloud Computing

The provision of on-demand access to computing resources and services hosted in distant data centres relies on networks. Some example of cloud computing is software as a service (SaaS), platform as a service (PaaS) and infrastructure as service (IaaS).

8. Online Banking and E-Commerce

Online banking and e-commerce platforms, where customers conduct financial transactions and make online purchases, require secure computer networks.

9. Enterprise Networks

In Computer networks, we have some networks that are only used in businesses and organizations so they can store data and share files and resources like printers, scanners, etc.

10. Healthcare

With the help of computer networks in the health industry, we can share patient records and store the records in the form of data that is easy and secure compared to the file method. Networks are also necessary for telemedicine and remote patient monitoring.

11. Education

Schools use networks to access online courses, virtual classrooms, and other online learning materials. Campuses of colleges and universities frequently have extensive computer networks.

12. Transportation and Logistics

The transportation sector uses Computer Networks to manage and track shipments, plan the best routes, and coordinate logistics activities.

13. Internet of Things (IoT) and Smart Homes

Through the Internet of Things (IoT), smart homes use networks to connect to and manage a variety of devices, including thermostats, security cameras, and smart appliances.

14. Scientific Research

To share data, work together on projects, and access high-performance computing resources for data analysis and scientific simulations, researchers use networks.

15. Government and Defense

With the help of computer networks, we can communicate, share data, and advance national defence. Government agencies and the military rely on secure networks.

Advantages of Computer Networking

Some of the main advantages of Computer Networking are discussed below:

- **Central Storage of Data:** Files can be stored on a central node (the file server) that can be shared and made available to every user in an organization.
- **Connectivity:** There is a negligible range of abilities required to connect to a modern computer network. The effortlessness of joining makes it workable for even youthful kids to start exploring the data.
- **Faster Problem-solving:** Since an extensive procedure is disintegrated into a few littler procedures and each is taken care of by all the associated gadgets, an explicit issue can be settled in lesser time.
- **Reliability:** Reliability implies backing up information. Due to some reason equipment crashes, and so on, the information gets undermined or inaccessible on one PC, and another duplicate of similar information is accessible on another workstation for future use, which prompts smooth working and further handling without interruption.
- **Flexible:** This innovation is known to be truly adaptable, as it offers clients the chance to investigate everything about fundamental things, for example, programming without influencing their usefulness.
- **Security through Authorization:** Security and protection of information are additionally settled through the system. As just the system clients are approved to get to specific records or applications, no other individual can crack the protection or security of information.
- **Storage capacity:** Since you will share data, records, and assets with other individuals, you need to guarantee all information and substance are legitimately put away in the framework. With this systems administration innovation, you can do most of this with no issue, while having all the space you require for capacity.

Disadvantages of Computer Networking

Some of the main disadvantages of Computer Networking are discussed below:

- **Expensive:** Execution of the network can be expensive in the case of an initial setup, as the wires and the cost of the cable are high and sometimes equipment is also costly.
- **Virus and Malware:** Computer Networking can lead to the spreading of viruses to another computer through the network.

- **Management of the network:** Management of the Network is quite difficult as it requires skilled persons to handle that large network. It requires training of people who are employed in this work.
- **Loss of Information:** In case of a crash of the Computer Network, it can lead to the loss of information or not being able to access information for some time.
- **The system can be Hacked:** In the case of Wide Area Networks(WAN), there is a threat of Hacking of the System. Some security features should be added to prevent such things.

Challenges of Computer Network

1. Performance Degradation –

Many time we have faced loss in data integrity and speed of a network which is generally down to poor transmissions and is also known as performance degradation. Every networks that may be large or small performance issue is everywhere but in large networks this performance degradation issue is high as communication has to be established with in a larger area and also by help of many network devices.

2. Security Issues –

When it comes to computer network there this security issue arises. It is one of top issue of computer network and a big challenge to network security engineers which generally involve protecting network from different cyber attacks, preventing unauthorized users to enter and access system, and maintaining network integrity. All these security issues increases with increase in network size when network size is large there chance of security issues are more.

3. Host Identification –

Small networks can be easily configured with help of manual addressing, but this becomes a serious problem in large networks when it comes to host identification. Because without any proper address of networking it becomes difficult to establish communication in network. So proper host identification is necessary for a network communication.

4. Configuration Conflicts –

Mainly large networks have to deal with configuration conflicts and busy networks, since a lot more traffic is going through it. But in small networks a couple of thousand IP addresses with unique host names are available so there is less chance of conflict in between devices. But now a days this problem is less as network structures are designed in such a way that deals with configuration conflicts.

5. Capacity Concern –

Now a days volume of data is so high which is produced from various sources. So network capacity also needs to grow with respect it. Today we are dealing with trends like Internet of Things (IoT), Big Data, Data science etc and next year something new will be also added so networks also need to improve capacity needs as well as facing cyber threats.

6. Slow Connectivity –

Slow connectivity over a network is more frustrating where a simple task takes a long time to be performed over network. It's often caused by large file transfers over a large area through network. It becomes an unwanted challenge for users when they work over computer network.

7. Monitoring and maintenance –

Monitoring and maintenance of global network is one of big challenge of current time. It becomes very difficult to monitor volume of a traffic in a large network.

Network Reference Models:

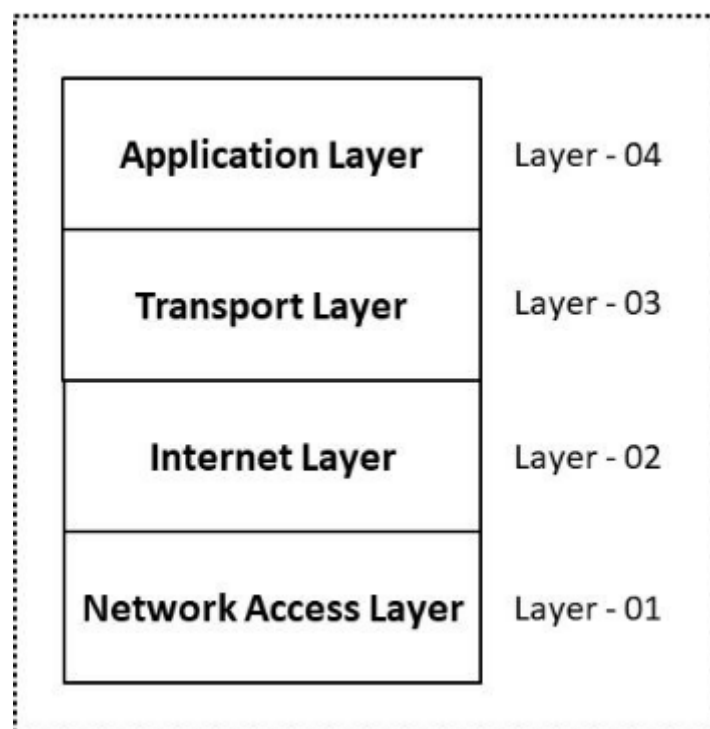
1. The OSI reference model

- OSI stands for Open System Interconnection is a reference model that describes how information from a software application in one computer moves through a physical medium to the software application in another computer.
- OSI consists of seven layers, and each layer performs a particular network function.
- OSI model was developed by the International Organization for Standardization (ISO) in 1984, and it is now considered as an architectural model for the inter-computer communications.
- OSI model divides the whole task into seven smaller and manageable tasks. Each layer is assigned a particular task.
- Each layer is self-contained, so that task assigned to each layer can be performed independently.

Layer 7	Application	Semantics Initiates a request or accepts a request
Layer 6	Presentation	Data Representations Adds formatting, display, and encryption information to the packet
Layer 5	Session	Dialog Coordination Adds traffic flow information to determine when the packet is sent
Layer 4	Transport	Reliable Transfer of Data Adds error-handling information
Layer 3	Network	Routing and Relaying Adds sequencing and address information to the packet
Layer 2	Data Link	Node-to-Node Data Transfer Adds error-checking information and prepares data for going on to the physical connection
Layer 1	Physical	Electrical and Optical Connection Sends packet as a bitstream

- **Physical Layer** – Its function is to transmit individual bits from one node to another over a physical medium.
- **Data Link Layer** – It is responsible for the reliable transfer of data frames from one node to another connected by the physical layer.
- **Network Layer** – It manages the delivery of individual data packets from source to destination through appropriate addressing and routing.
- **Transport Layer** – It is responsible for delivery of the entire message from the source host to destination host.
- **Session Layer** – It establishes sessions between users and offers services like dialog control and synchronization.
- **Presentation Layer** – It monitors syntax and semantics of transmitted information through translation, compression, and encryption.
- **Application Layer** – It provides high-level APIs (application program interface) to the users.

2. The TCP/IP Four-Layer Reference Model



1. Network Access Layer

It is a group of applications requiring network communications. This layer is responsible for generating the data and requesting connections. It acts on behalf of the sender and the Network Access layer on the behalf of the receiver. During this article, we will be talking on the behalf of the receiver.

The packet's network protocol type, in this case, TCP/IP, is identified by network access layer. Error prevention and "framing" are also provided by this layer. [Point-to-Point Protocol \(PPP\)](#) framing and Ethernet IEEE 802.2 framing are two examples of data-link layer protocols.

2. Internet Layer

This layer parallels the functions of OSI's Network layer. It defines the protocols which are responsible for the logical transmission of data over the entire network. The main protocols residing at this layer are as follows: IP, ICMP, ARP

The Internet Layer is a layer in the Internet Protocol (IP) suite, which is the set of protocols that define the Internet. The Internet Layer is responsible for routing packets of data from one device to another across a network. It does this by assigning each device a unique IP address, which is used to identify the device and determine the route that packets should take to reach it.

3. Transport Layer

The TCP/IP transport layer protocols exchange data receipt acknowledgments and retransmit missing packets to ensure that packets arrive in order and without error. End-to-end communication is referred to as such. Transmission Control Protocol (TCP) and User Datagram Protocol are transport layer protocols at this level (UDP).

4. Application Layer

This layer is analogous to the transport layer of the OSI model. It is responsible for end-to-end communication and error-free delivery of data. It shields the upper-layer applications from the complexities of data. The three main protocols present in this layer are:

- HTTP and HTTPS: [HTTP](#) stands for Hypertext transfer protocol. It is used by the World Wide Web to manage communications between web browsers and servers. HTTPS stands for HTTP-Secure. It is a combination of HTTP with SSL (Secure Socket Layer). It is efficient in cases where the browser needs to fill out forms, sign in, authenticate, and carry out bank transactions.
- SSH: [SSH](#) stands for Secure Shell. It is terminal emulations software similar to Telnet. The reason SSH is preferred is because of its ability to maintain the encrypted connection. It sets up a secure session over a TCP/IP connection.

Difference between TCP/IP and OSI Model

TCP/IP	OSI
TCP refers to Transmission Control Protocol.	OSI refers to Open Systems Interconnection.
TCP/IP uses both the session and presentation layer in the application layer itself.	OSI uses different session and presentation layers.

TCP/IP	OSI
TCP/IP follows connectionless a horizontal approach.	OSI follows a vertical approach.
The Transport layer in TCP/IP does not provide assurance delivery of packets.	In the OSI model, the transport layer provides assurance delivery of packets.
Protocols cannot be replaced easily in TCP/IP model.	While in the OSI model, Protocols are better covered and are easy to replace with the technology change.
TCP/IP model network layer only provides connectionless (IP) services. The transport layer (TCP) provides connections.	Connectionless and connection-oriented services are provided by the network layer in the OSI model.

Network Devices:

1. Repeater –

A repeater operates at the physical layer. Its job is to regenerate the signal over the same network before the signal becomes too weak.

- A repeater connects different segments of a LAN
- A repeater forwards every frame it receives
- A repeater is a regenerator, not an amplifier
- It can be used to create a single extended LAN

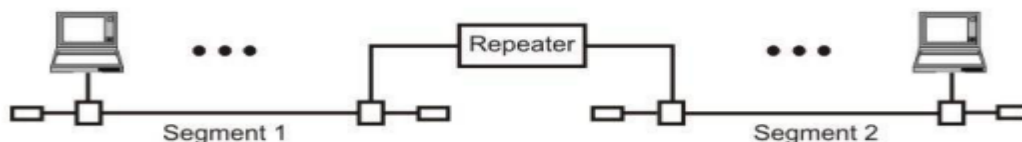
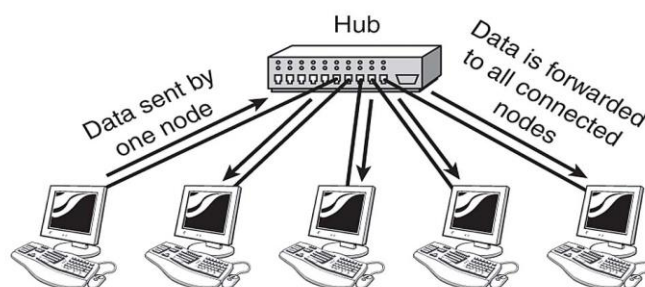


Figure Repeater connecting two LAN segments

2. HUB –



A hub is a basically multi-port repeater. A hub connects multiple wires coming from different branches, for example, the connector in star topology which connects different stations. Hubs cannot filter data, so data packets are sent to all connected devices.

Types of Hub

- **Active Hub:** - These are the hubs that have their power supply and can clean, boost, and relay the signal along with the network. It serves both as a repeater as well as a wiring center. These are used to extend the maximum distance between nodes.
- **Passive Hub:** - These are the hubs that collect wiring from nodes and power supply from the active hub. These hubs relay signals onto the network without cleaning and boosting them and can't be used to extend the distance between nodes.
- **Intelligent Hub:** - It works like an active hub and includes remote management capabilities. They also provide flexible data rates to network devices. It also enables an administrator to monitor the traffic passing through the hub and to configure each port in the hub.

3. Bridge –

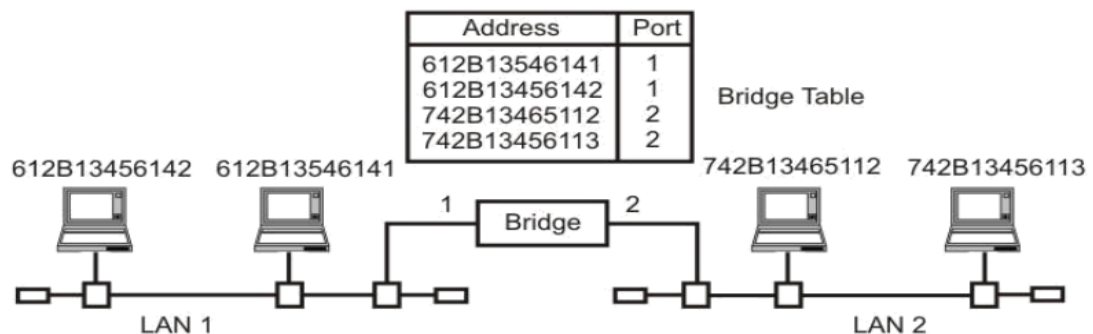


Figure A bridge connecting two separate LANs

A bridge operates at the data link layer. A bridge is a repeater; with add on the functionality of filtering content by reading the MAC addresses of the source and destination. It is also used for interconnecting two LANs working on the same protocol. It has a single input and single output port, thus making it a 2 port device.

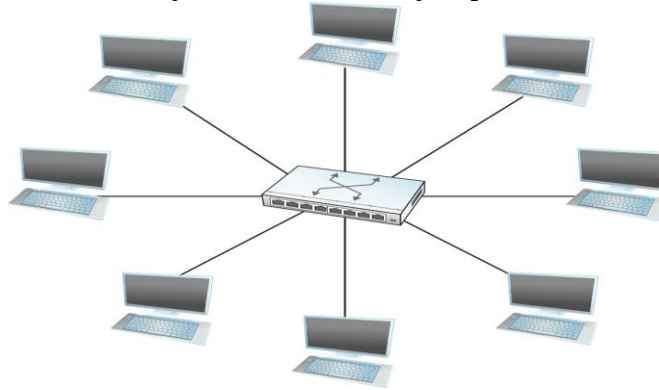
- A bridge operates both in physical and data-link layer
- A bridge uses a table for filtering/routing
- A bridge does not change the physical (MAC) addresses in a frame
- Types of bridges: Transparent Bridges ,Source routing bridges

4. Switch –

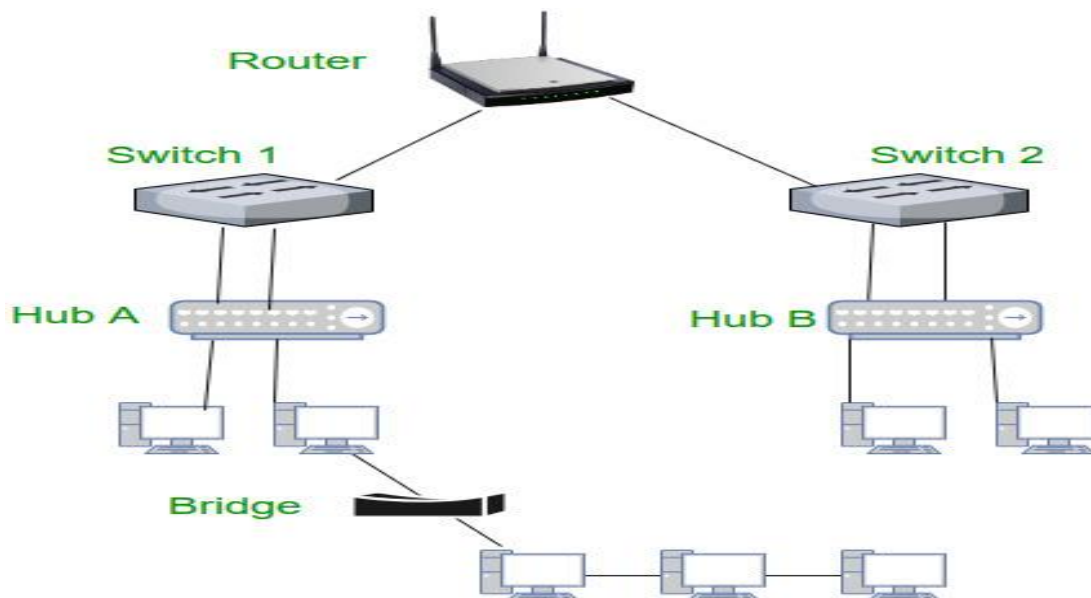
A switch is a multi-port bridge with a buffer and a design that can boost its efficiency (large number of ports imply less traffic) and performance. Switch is data link layer device. Switch can perform error checking before forwarding data, that makes it very efficient as it does not forward packets that have errors and forward good packets selectively to correct port only

A switch is essentially a fast bridge having additional sophistication that allows faster processing of frames. Some of important functionalities are:

- Ports are provided with buffer
- Switch maintains a directory: #address - port#
- Each frame is forwarded after examining the #address and forwarded to the proper port# Three possible forwarding approaches: Cut-through, Collision-free and Fully buffered as briefly explained below.



5. Routers – A router is a device like a switch that routes data packets based on their IP addresses. The router is mainly a Network Layer device. Routers normally connect LANs and WANs and have a dynamically updating routing table based on which they make decisions on routing the data packets. The router divides the broadcast domains of hosts connected through it.



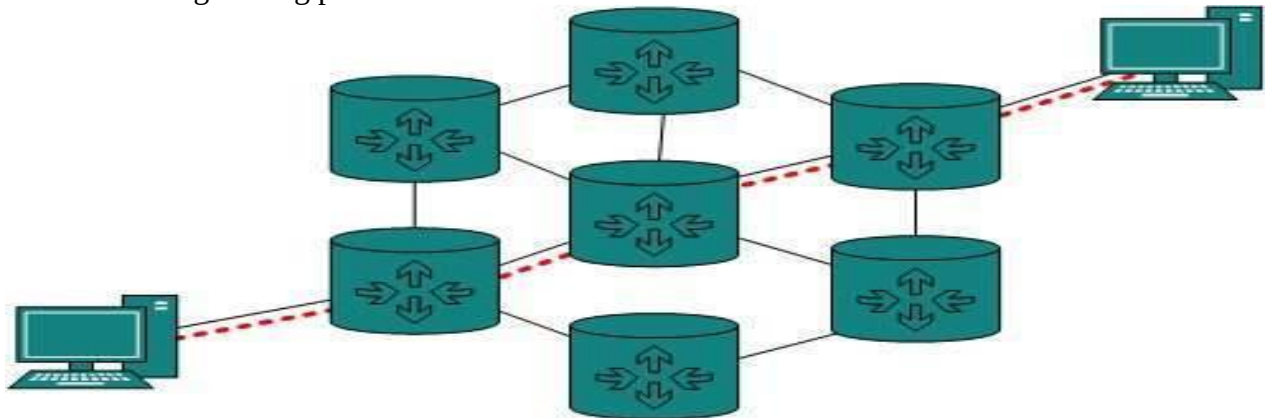
6. Gateway – A gateway, as the name suggests, is a passage to connect two networks that may work upon different networking models. They work as messenger agents that take data from one system, interpret it, and transfer it to another system. Gateways are also called protocol converters and can operate at any network layer. Gateways are generally more complex than switches or routers. A gateway is also called a protocol converter.

Introduction to Internetworking.

In real world scenario, networks under same administration are generally scattered geographically. There may exist requirement of connecting two different networks of same kind as well as of different kinds. Routing between two networks is called internetworking.

Networks can be considered different based on various parameters such as, Protocol, topology, Layer-2 network and addressing scheme.

In internetworking, routers have knowledge of each other's address and addresses beyond them. They can be statically configured go on different network or they can learn by using internetworking routing protocol.

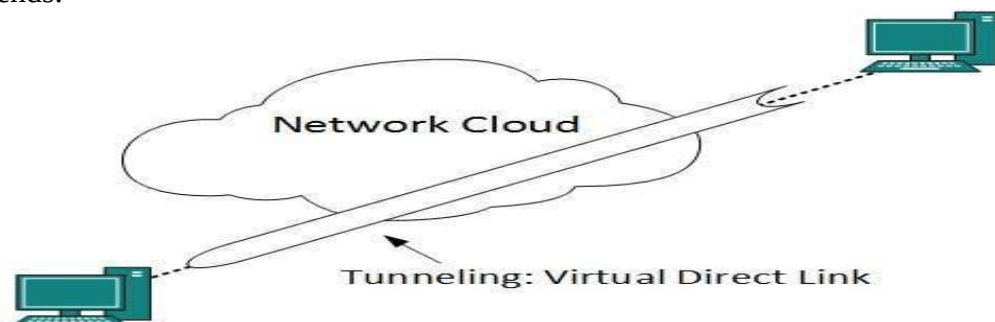


Routing protocols which are used within an organization or administration are called Interior Gateway Protocols or IGP. RIP, OSPF are examples of IGP. Routing between different organizations or administrations may have Exterior Gateway Protocol, and there is only one EGP i.e. Border Gateway Protocol.

Tunneling

If they are two geographically separate networks, which want to communicate with each other, they may deploy a dedicated line between or they have to pass their data through intermediate networks.

Tunneling is a mechanism by which two or more same networks communicate with each other, by passing intermediate networking complexities. Tunneling is configured at both ends.



When the data enters from one end of Tunnel, it is tagged. This tagged data is then routed inside the intermediate or transit network to reach the other end of Tunnel. When data exists the Tunnel its tag is removed and delivered to the other part of the network.

Both ends seem as if they are directly connected and tagging makes data travel through transit network without any modifications.